ative ring

atrices:

$$\begin{pmatrix} 2 & 3 \\ -1 & 4 \end{pmatrix}\begin{pmatrix} 1 & 6 \\ 5 & -1 \end{pmatrix} = \begin{pmatrix} 17 & 9 \\ 19 & -10 \end{pmatrix}$$

$$\begin{matrix} 3 \\ 4 \end{pmatrix} + \begin{pmatrix} 1 & 6 \\ 5 & -1 \end{pmatrix} = \begin{pmatrix} 3 & 9 \\ 4 & 3 \end{pmatrix}$$

$$0 \approx \begin{pmatrix} 0 & 0 \\ 0 & 0 \end{pmatrix} \quad -1 \approx \begin{pmatrix} -1 & 0 \\ 0 & -1 \end{pmatrix}$$ always

since multiplicative inverses don't exist

$M \neq 0)$

## Other examples of fields

Complex rationals:

$a + bi, \quad a, b \in \mathbb{Q}$

$$(2+3i) + \left(5 - \tfrac{3}{2}i\right) = 7 + \tfrac{3}{2}i$$

$$(2+3i)(1-i) = 5 + i \quad (i^2 = -1)$$

$$\frac{1}{(2+3i)}\frac{2-3i}{(2-3i)} = \frac{2-3i}{2^2+3^2} = \frac{2}{13} - \frac{3}{13}i$$